Simulation-Based Wrist Exoskeleton Assistance after Stroke: Optimal Control vs. Reinforcement Learning

P2: Literature Map, Method Outline and Baseline Simulation

Team Members:

Vedant Joshi

Navya Nagendra

Harshil Safi

Guided by:
Prof. Hamidreza Heidari

P2 at a Glance:

TECHNISCHE HOCHSCHULE DEGGENDORF THD

1 **Literature map** — Done

20 sources in 4 clusters: clinical problem, wrist devices, musculoskeletal optimal control, learning-based control

2 **Method outline** — Done

Two-route pipeline (Moco plan and MyoSuite RL), KPIs K1–K4 made measurable

3 **Baseline simulation** — Done

MoBL-ARMS cup lift-and-hold converges; the six wrist muscles do all the work

4 **Bibliography** — Done

All references checked online; list on the last slides

0.02 %

peak reserve torque vs. gravity torque in the baseline (limit 10 %)

≈ 10 %

ECRB / ECRL activation that holds the 0.3 kg cup (MoBL-ARMS)

≈ 10.7 %

extensor strength below which the task fails (K1 preview)

Baseline: MoBL-ARMS. K1 preview: planar cross-check model.

Literature Map:

Clinical problem (why)

- Nijland 2010: early finger extension predicts arm recovery after stroke
- Wrist drop: weak extensors, hand falls under gravity

Wrist assist devices (what exists)

- Hussain 2020: review of wrist rehabilitation robots
- Lambelet 2017, 2020: eWrist, 1-DoF wrist exo, < 500 g
- KhalilianMotamed Bonab 2024: exosuit designed in OpenSim

Our study

Size wrist assistance for a weakened wrist (spring or motor?) and compare a Moco plan with an RL policy

Optimal control (Route A)

- Delp 2007, Seth 2018: OpenSim
- Dembia 2020: OpenSim Moco
- Holzbaur 2005, Saul 2015, McFarland 2019: MoBL-ARMS
- De Groote 2016: muscle model

Learning-based control (Route B)

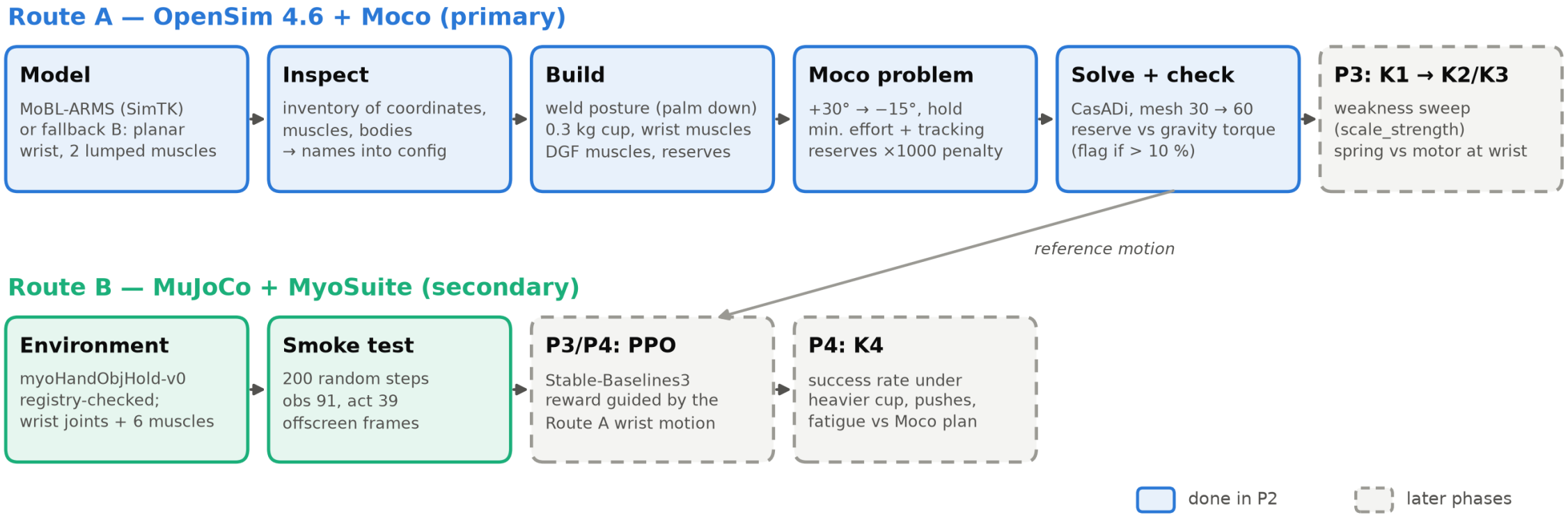
- Todorov 2012: MuJoCo
- Caggiano 2022, Wang 2022: MyoSuite, MyoSim
- Schulman 2017: PPO
- Luo 2024, Chavarrías 2025: RL exo control

Gap: no study we found sizes wrist assistance for a weakened post-stroke wrist, or compares an optimal plan with a learned controller.

From Literature to Our Pipeline:

Source	What it gives us	Where we use it
Holzbaur 2005; Saul 2015; McFarland 2019	MoBL-ARMS upper-limb model (OpenSim 4.1 version)	Route A model for the baseline
Dembia 2020 (Moco)	Direct-collocation optimal control	Baseline solve, K1–K3 studies
De Groote 2016	Fast, robust muscle formulation	DeGrooteFregly muscles in every solve
Wang 2022; Caggiano 2022	MyoHand model (from MoBL-ARMS) and MyoSuite tasks	Route B environment; planar cross-check model
Lambelet 2017, 2020; Hussain 2020	A 1-DoF wrist flexion/extension exo is wearable and realistic	Assist modelled as torque at the wrist axis
Schulman 2017; Raffin 2021	PPO and Stable-Baselines3	Route B policy training (P3/P4)

Method Outline:

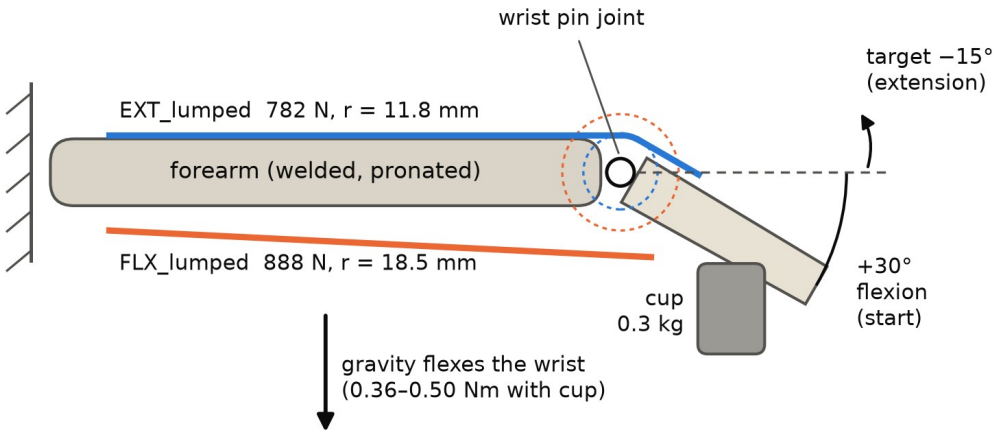


Solid boxes: built and tested in P2. Dashed boxes: P3/P4. The Moco wrist motion becomes the reference for the RL reward.

Model and Task: MoBL-ARMS:

- MoBL-ARMS upper limb (OpenSim 4.1 version, torso fixed)
- Arm at the side, elbow 90°, forearm pronated (palm down): gravity flexes the wrist
- Wrist flexion free; deviation braced at 0°; 9 joints welded
- 6 wrist muscles: ECRL, ECRB, ECU, FCR, FCU, PL (DeGroote–Fregly, rigid tendon)
- 0.3 kg cup in the hand: gravity torque 0.37–0.54 Nm

Model fixes we needed: exact-pose welds, a frozen moving path point (FCU) and a deviation brace; all checked in tests.



Task schematic, drawn on the planar cross-check model (two lumped muscles) used to verify the pipeline.

Moco Problem Formulation:

1 Task

- Wrist from +30° (dropped) to −15° (extended)
- 1.0 s; last 30 % is a hold
- Zero velocity at start and end

2 Objective

- Minimise muscle effort (Σ excitation²)
- Reserve torque penalised ×1000
- Soft tracking of a smooth move-then-hold path

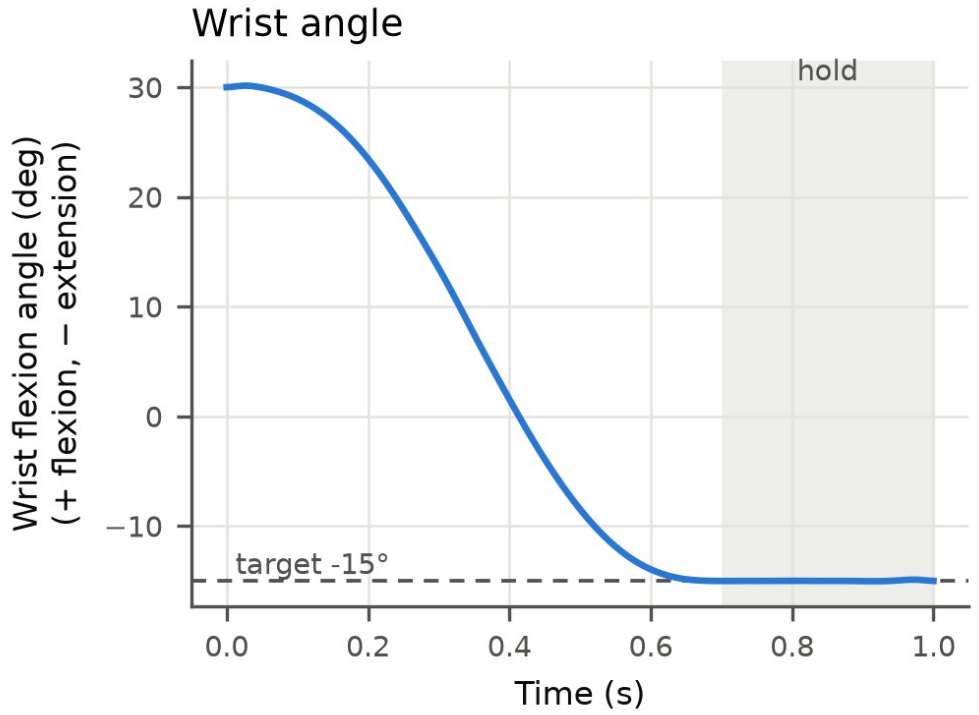
3 Pass / fail checks

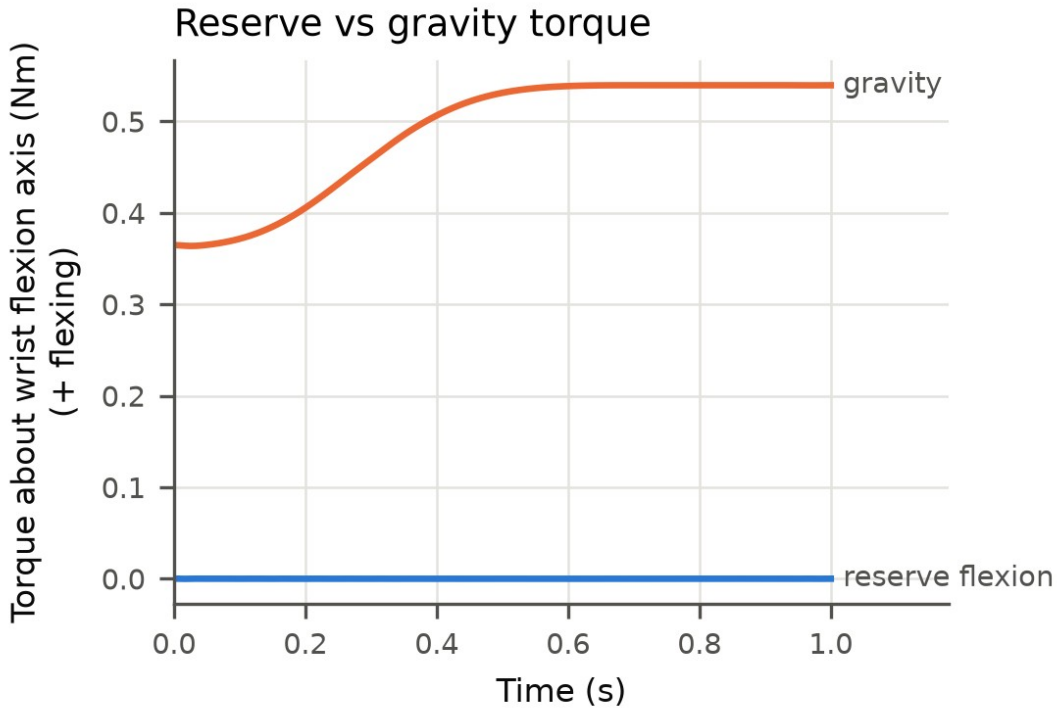
- Reserve ≤ 10 % of gravity torque
- Hold drift ≤ 2°
- Solver must converge

4 Solver

- Moco CasADi + IPOPT, Hermite–Simpson
- Coarse 30 → refined 60 mesh intervals
- Tolerances 1e-3

Baseline Result (100 % Strength):





Converged

both meshes; refined: 233 iterations, 28 min

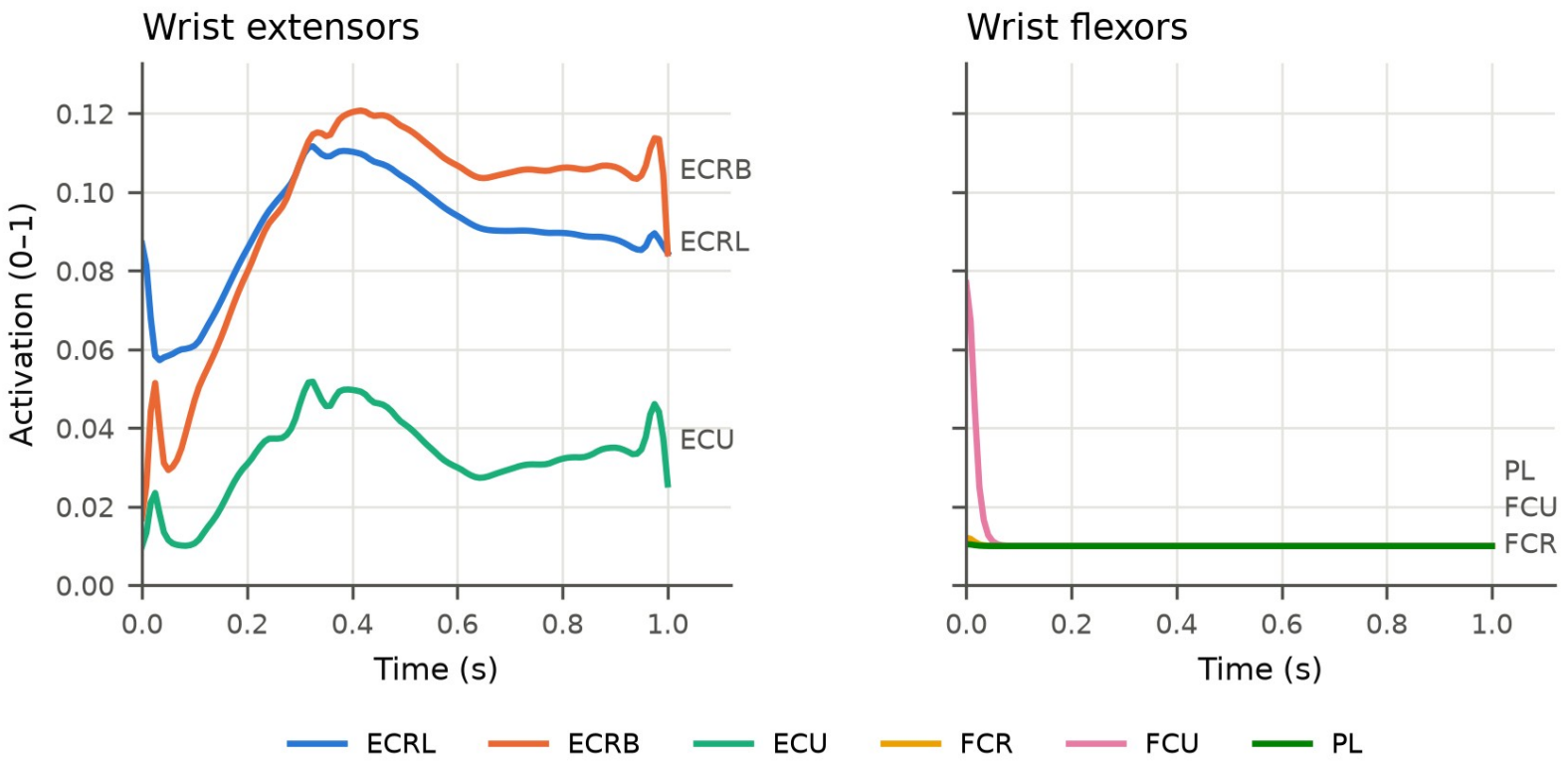
−15.00°

target reached; held within 0.11°

0.02 %

peak reserve / gravity torque: muscles do the work

Muscle Activity:



≈ 10 %

ECRB / ECRL activation holds the 0.3 kg cup level

0.01

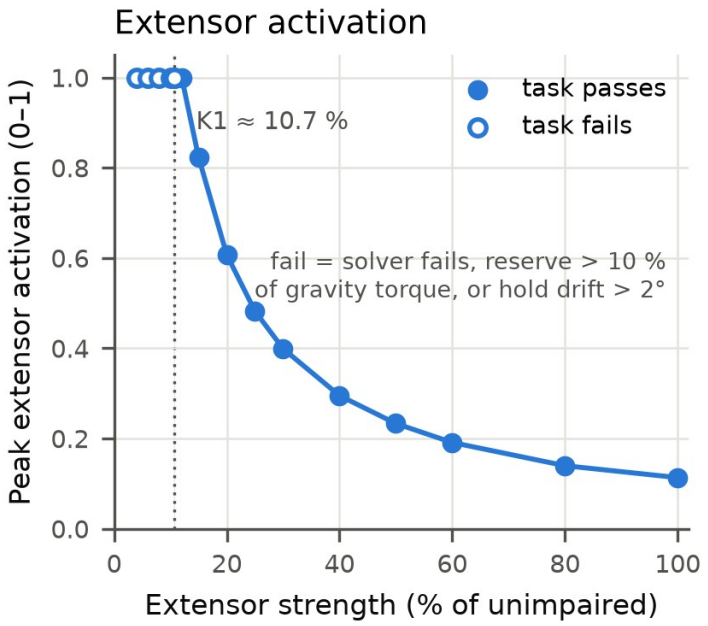
FCR, FCU, PL activation (model minimum)

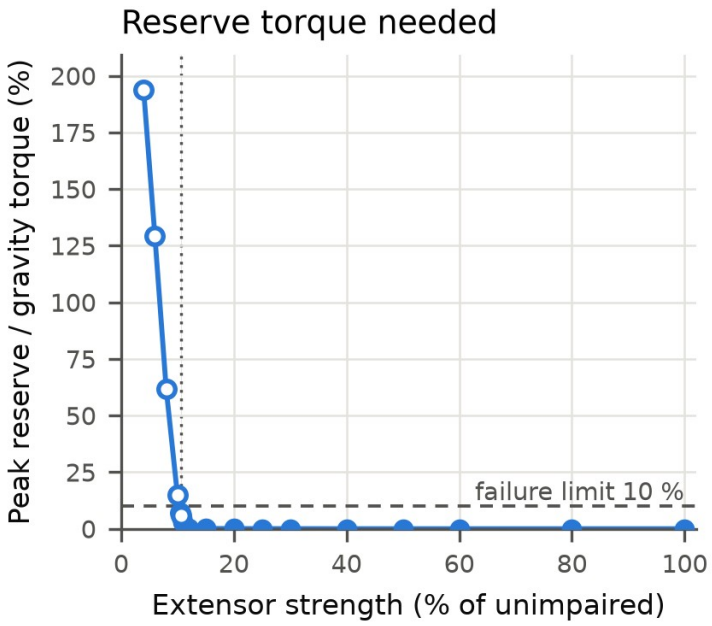
Starting point for K1: how much weaker can the extensor get?

ECRB and ECRL do most of the work, ECU helps; the flexors stay at their minimum activation.

K1 Preview: Weakness Sweep:

Fallback B: planar 2-muscle wrist model | preliminary P3 result





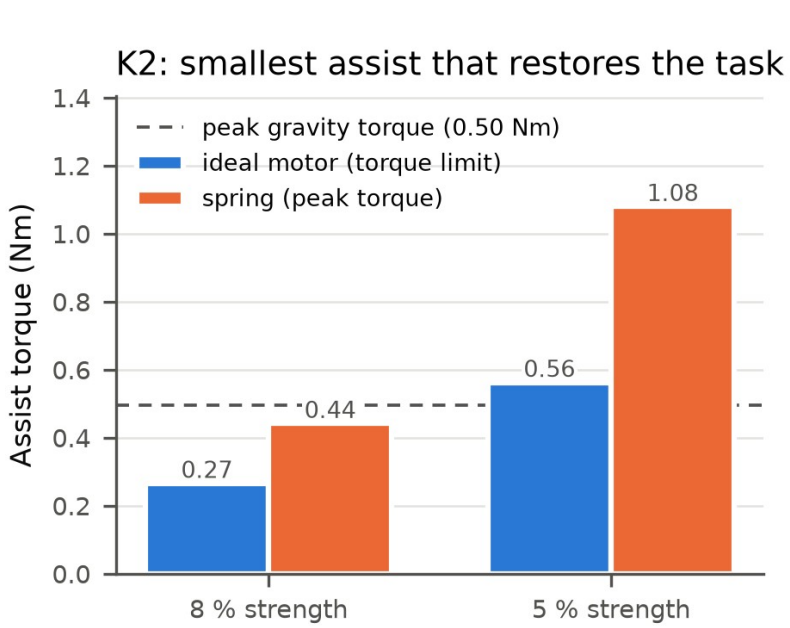
≈ 10.7 %

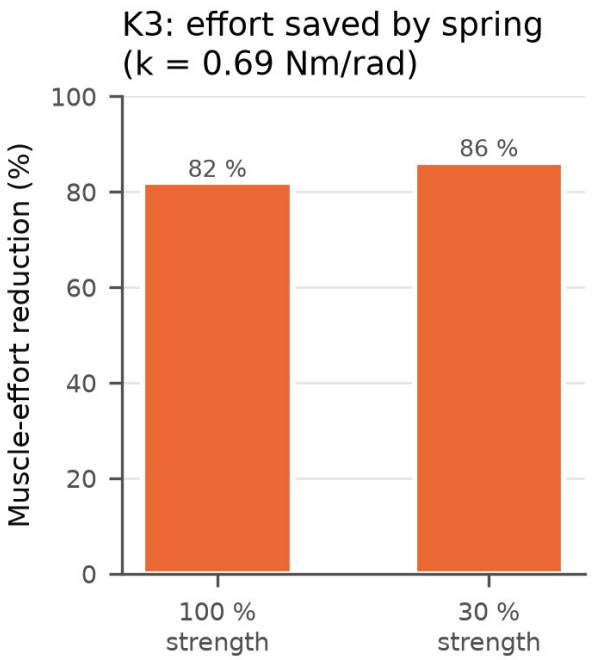
of unimpaired extensor strength: below this the cup task fails

- Activation saturates at 1.0 below ~12 %
- Threshold found by bisection: 10.62–10.75 %
- Preview on the planar cross-check model; final K1 on MoBL-ARMS in P3

K2/K3 Preview: Motor vs. Spring:

Fallback B: planar 2-muscle wrist model | preliminary P3 result





0.27 Nm

motor that restores the task at 8 % strength (0.56 Nm at 5 %)

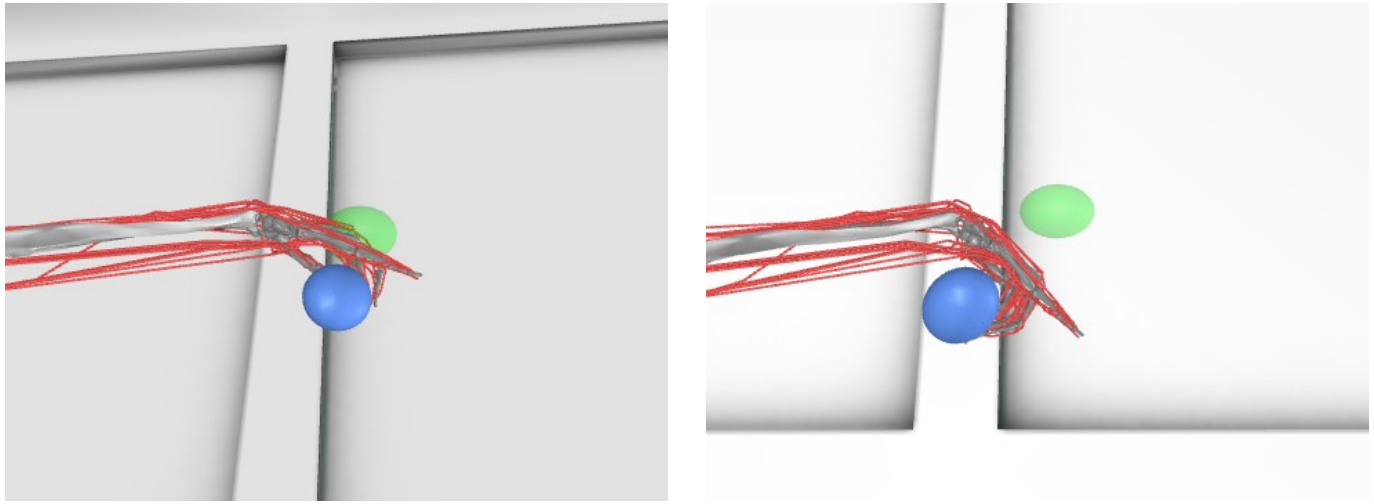
1.6–1.9×

spring peak torque vs. motor: a spring is enough, but must be stronger

82–86 %

less muscle effort with the spring when the task still works

Route B: MyoSuite Environment:



myoHandObjHoldFixed-v0 under random actions (offscreen render).

Check	Result
Registry	226 myo environments, 24 myoHand
Obs. / action	91 / 39 (muscles)
Wrist joints	flexion, deviation, pro_sup
Wrist muscles	ECRL, ECRB, ECU, FCR, FCU, PL
Smoke test	200 random steps, no errors

Route B plan (P3/P4)

- PPO (Stable-Baselines3); reward tracks the Moco wrist motion, keeps the cup level and penalises effort
- Fingers held in a fixed grasp first; policy outputs muscle (and device) commands
- K4: compare the RL policy with the Moco plan under a heavier cup, pushes and fatigue

Limitations and Validity:

Limitation	Effect	Plan
K1–K3 previews on the planar model	Thresholds are model-specific	Re-run on MoBL-ARMS (≈45 min per solve)
Wrist deviation braced at 0°	Brace takes radial side effects (0.49 Nm)	Test with deviation free
Assumed fibre and tendon lengths	K1 value may shift	Sensitivity analysis in P3
Hold enforced by a soft tracking goal	Kinematics partly prescribed	Compare with a purely predictive variant
Spring rest angle and motor weight fixed	K2/K3 depend on these choices	Optimise them as Moco parameters
Generic adult, one 1 s task	Not a clinical prediction	State as relative findings (P1 ethics)

Next Steps (P3):

1

K1 on MoBL-ARMS

coarse-mesh sweep with warm starts

P3 · 18 Nov

2

Sensitivity

cup mass, speed, deviation free

P3 · 18 Nov

3

Spring vs. motor

stiffness and rest angle as Moco parameters

P3 · 2 Dec

4

Route B training

PPO in MyoSuite, Moco-guided reward

Weeks 48–50

5

Robustness (K4)

heavier cup, pushes, fatigue

P4 · 16 Dec

Already in place: MoBL-ARMS baseline, sweep and assist scripts, tests and figures; P3 re-runs them on MoBL-ARMS.

Bibliography (1/2):

- Nijland, R. H. M., et al. (2010). Presence of finger extension and shoulder abduction within 72 hours after stroke predicts functional recovery: the EPOS cohort study. Stroke, 41(4), 745–750.
- Hussain, S., Jamwal, P. K., Van Vliet, P., & Ghayesh, M. H. (2020). State-of-the-art robotic devices for wrist rehabilitation: Design and control aspects. IEEE Transactions on Human-Machine Systems, 50(5), 361–372.
- Lambelet, C., Lyu, M., Woolley, D., Gassert, R., & Wenderoth, N. (2017). The eWrist – A wearable wrist exoskeleton with sEMG-based force control for stroke rehabilitation. Proc. IEEE ICORR, 726–733.
- Lambelet, C., et al. (2020). Characterization and wearability evaluation of a fully portable wrist exoskeleton for unsupervised training after stroke. Journal of NeuroEngineering and Rehabilitation, 17, 132.
- KhalilianMotamed Bonab, A., Chiaradia, D., Frisoli, A., & Leonardis, D. (2024). A framework for modeling, optimization, and musculoskeletal simulation of an elbow–wrist exosuit. Robotics, 13(4), 60.
- Luo, S., et al. (2024). Experiment-free exoskeleton assistance via learning in simulation. Nature, 630, 353–359.
- Delp, S. L., et al. (2007). OpenSim: Open-source software to create and analyze dynamic simulations of movement. IEEE TBME, 54(11).
- Seth, A., et al. (2018). OpenSim: Simulating musculoskeletal dynamics and neuromuscular control. PLOS Computational Biology, 14(7).
- Dembia, C. L., et al. (2020). OpenSim Moco: Musculoskeletal optimal control. PLOS Computational Biology, 16(12), e1008493.
- Holzbaur, K. R. S., Murray, W. M., & Delp, S. L. (2005). A model of the upper extremity for simulating musculoskeletal surgery and analyzing neuromuscular control. Annals of Biomedical Engineering, 33(6), 829–840.

Bibliography (2/2):

- Saul, K. R., et al. (2015). Benchmarking of dynamic simulation predictions in two software platforms using an upper limb musculoskeletal model. Computer Methods in Biomechanics and Biomedical Engineering.
- McFarland, D. C., McCain, E. M., Poppo, M. N., & Saul, K. R. (2019). Spatial dependency of glenohumeral joint stability during dynamic unimanual and bimanual pushing and pulling. Journal of Biomechanical Engineering. doi:10.1115/1.4043035
- De Groote, F., Kinney, A. L., Rao, A. V., & Fregly, B. J. (2016). Evaluation of direct collocation optimal control problem formulations for solving the muscle redundancy problem. Annals of Biomedical Engineering. doi:10.1007/s10439-016-1591-9
- Winter, D. A. (2009). Biomechanics and Motor Control of Human Movement (4th ed.). Wiley.
- Todorov, E., Erez, T., & Tassa, Y. (2012). MuJoCo: A physics engine for model-based control. Proc. IEEE/RSJ IROS, 5026–5033.
- Caggiano, V., et al. (2022). MyoSuite: A contact-rich simulation suite for musculoskeletal motor control. L4DC, PMLR 168.
- Wang, H., Caggiano, V., Durandau, G., Sartori, M., & Kumar, V. (2022). MyoSim: Fast and physiologically realistic MuJoCo models for musculoskeletal and exoskeletal studies. Proc. IEEE ICRA, 8104–8111.
- Schulman, J., Wolski, F., Dhariwal, P., Radford, A., & Klimov, O. (2017). Proximal policy optimization algorithms. arXiv:1707.06347.
- Raffin, A., et al. (2021). Stable-Baselines3: Reliable reinforcement learning implementations. Journal of Machine Learning Research, 22(268), 1–8.
- Chavarrías, A., Rodriguez-Cianca, D., & Lanillos, P. (2025). Adaptive torque control of exoskeletons under spasticity conditions via reinforcement learning. arXiv:2503.11433.

Thank You

Questions?

Repository:
github.com/Rationaliser/wrist-exo-assist-sim